



Project: Rocketry Training - Aerospace Simulation Workshop Project

FEM + CFD Project using SimScale

Project Submission Guidelines

- Submit the project report in PDF format.
- Include screenshots of simulations and results.
- Clearly mention assumptions used in the simulations.
- Explain observations and engineering conclusions properly.

Project Evaluation Criteria

Criteria	Weightage
Understanding of Concepts	20%
Simulation Setup	20%
Result Interpretation	25%
Engineering Reasoning	25%
Report Presentation	10%

PART A — THEORETICAL SECTION

Q1. Explain Finite Element Method (FEM) in your own words.

Discuss what FEM is, why it is used, and its applications in aerospace engineering.

Q2. What is Von Mises Stress and why is it important?

Explain yielding and structural failure concepts.

Q3. Explain the role of meshing in FEM simulations.

Discuss accuracy, computational cost, and mesh dependency.

Q4. Differentiate between CFD and FEM.

Compare objectives, outputs, and engineering applications.

Q5. Why are boundary conditions important in simulations?

Explain supports, loads, constraints, and incorrect assumptions.

PART B — FEM PROJECT

TASK 1 — Rocket Fin Geometry & Material



Create/import a rocket fin geometry and assign material properties.

TASK 2 — Boundary Conditions

Apply realistic fixed supports and aerodynamic loads.

TASK 3 — Mesh Study

Perform coarse, medium, and fine mesh analysis and compare results.

TASK 4 — Result Interpretation

Analyze Von Mises stress, deformation, and critical regions.

TASK 5 — Factor of Safety

Calculate factor of safety and determine structural safety.

PART C — CFD PROJECT

TASK 1 — Flow Domain Creation

Create the external flow region around the geometry.

TASK 2 — CFD Boundary Conditions

Apply inlet velocity, outlet pressure, and wall conditions.

TASK 3 — CFD Mesh Generation

Generate mesh and explain refinement strategy.

TASK 4 — Flow Visualization

Analyze pressure contours, velocity contours, and streamlines.

TASK 5 — Aerodynamic Discussion

Discuss drag, wake formation, and aerodynamic stability.

BONUS PROJECT TASK — Design Optimization

Modify the fin design to reduce stress or aerodynamic drag. Compare the original and optimized design using proper engineering observations.



Recommended Report Structure

1. Aim
2. Theory
3. Geometry
4. Material Properties
5. Boundary Conditions
6. Mesh Details
7. Results
8. Discussion
9. Conclusion
10. References

Final Note:

Simulation results are highly dependent on assumptions, mesh quality, material properties, and boundary conditions. Engineering judgment must always accompany simulation results.

